

David Chao

Software Engineer | Backend Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience building scalable services, APIs, and developer tools in Kotlin, Java, and Python for production systems. Led initiatives that improved reliability, accelerated deployments, and expanded system observability across Kubernetes-based infrastructure. Strong focus on backend architecture, performance, reliability, distributed systems, and enabling teams through data-driven tooling and automation.

PROFESSIONAL EXPERIENCE

Software Engineer II, Expedia

August 2022 - May 2025

- Secured 99.9% uptime for critical services by spearheading the implementation of 40+ Availability SLOs, building 10+ reliability dashboards, and presenting weekly operational analytics to VPs to drive cross-team accountability
- Reduced incident resolution time by 50% as part of support rotation by proactively monitoring availability SLOs, using Datadog dashboard and alerts to detect degradations and notify 5+ product teams before incidents escalated
- Advanced company-wide 15-minute recovery targets by optimizing Spinnaker CI/CD pipelines, assisting 5+ teams in adopting progressive deployments, and tuning Kubernetes configurations to support faster fault recovery
- Shortened load test setup time by 90% by independently designing and developing a React frontend and Kotlin backend tool to streamline performance test preparation for product teams
- Boosted limited clickstream visibility by owning Clicker, a Kotlin service that streams Kafka metrics to Datadog, enhancing it with regex filtering, improved test coverage, and onboarding documentation for external teams
- Improved Blazemeter load test accuracy by refining a preexisting Python-based traffic prediction model through updates to dataframes, cron schedules, and adjustments to its linear regression and Fourier series components
- Introduced frontend observability by leading the web implementation of a multi-platform Real User Monitoring(RUM) initiative, instrumenting 20+ React components to enable proactive issue detection in Datadog
- Strengthened incident readiness by leading development of an internal SLO playground app that deepened the team's understanding of Expedia's tech, architecture, and monitoring practices

Software Developer, General Motors

July 2020 - July 2022

- Modernized GM's customer-facing owner center app by migrating 5+ AngularJS pages to Angular, improving maintainability ahead of AngularJS end-of-life in 2021
- Delivered secure, scalable data across GM systems by building 5+ Java REST APIs and refactoring data access for a stateless architecture
- Achieved PCI compliance and PII protection by fixing 100+ OSS and static code vulnerabilities using Black Duck and Fortify
- Won GM Austin 2020 hackathon by developing a contactless Android car demo app for executive use

Software Engineer Summer Internship, WellSky

June 2019 - August 2019

- Built a DevOps Dashboard in Angular to surface pipeline health, reducing issue resolution time by 10%
- Created over 50 unit and integration tests to help the EMR system meet HIPAA compliance for data handling
- Prototyped AWS-based IoT infrastructure to explore remote patient monitoring solutions

SKILLS AND TECHNOLOGIES

Languages: Java • Python • Kotlin • JavaScript • TypeScript • C++ • C • SQL

Framework & Libraries: Node.js • React • GraphQL • gRPC • REST APIs • Software Architecture • NodeJs • Angular • Git • Github • Spring • Spring MVC

Tools & DevOps: Kubernetes • Docker • Datadog • CI/CD Pipelines • Copilot • Splunk • Catchpoint • Tableau • Kafka • Jenkins • Spinnaker • BlazeMeter • JMeter • Selenium • Maven • Jira • Swagger • Fortify • Black Duck OSS

Cloud Platform: AWS

Methodologies: Agile • Scrum • API Development • Observability • Unit Testing

EDUCATION

Bachelor of Science, Electrical and Computer Engineering, May 2020

The University of Texas at Austin